Traffic Signal Control with Cycle-Level DQN

Author: Snehita Satya Sri Kandula, Alejandra Rivas Montes, Naiji Zhai

(Authors listed alphabetically by surname)

Abstract

This paper presents a Deep Q-Network (DQN) approach to adaptive traffic signal control at a single intersection. The agent operates at cycle-level granularity, adjusting green-phase durations at each complete signal cycle rather than switching phases at every simulation step. The environment is a custom Gymnasium simulation with stochastic, time-varying vehicle arrivals and a four-phase fixed-cycle structure, modeled after Moreno-Malo et al. (2024). While the original work evaluates a multi-intersection SUMO environment, this project focuses on a simplified single-intersection setting to isolate learning dynamics. The DQN agent is compared against six baseline policies including fixed-time, demand-aware, tuned-static, queue-based heuristic, and random policies. A series of ablation experiments examines the effects of training duration, network size, and reward shaping on agent performance under both symmetric and asymmetric-peaky demand conditions. The strongest result is observed under asymmetric-peaky demand, where DQN achieves a reward gain of 706 points and a 7.741% reduction in average waiting time compared to the best fixed-time baseline. Under symmetric demand, gains are smaller but consistently positive under stable hyperparameter settings, yielding approximately 1.0% to 1.3% wait reduction.

1. Introduction

Traffic signal control is a sequential decision problem: each signal decision changes future queues, delays, and discharge opportunities. Traditional fixed-time control assigns predetermined green durations to each phase regardless of current traffic conditions. While fixed-time policies are simple, predictable, and robust, they cannot adapt to directional imbalance or time-varying demand. In settings where one approach receives consistently higher traffic volume, a fixed policy wastes green time on empty lanes while congested approaches wait unnecessarily.

Reinforcement learning provides a natural framework for adaptive traffic signal control. An agent interacts with a traffic environment, observes the current state of queues and flows, and selects timing actions to minimize congestion and waiting time. Deep Q-Networks (DQN), introduced by Mnih et al. (2015) [2], combine Q-learning with deep neural networks, experience replay, and a target network to stabilize training. Their work demonstrated human-level control on Atari games and established DQN as a foundational algorithm in deep reinforcement learning.

Moreno-Malo et al. (2024) [1] demonstrated a distributed DQN agent system for traffic signal control at intersections simulated in SUMO, achieving a 44% reduction in waiting times compared to fixed-time control. Their control structure uses a fixed four-phase cycle, cycle-level timing adjustment actions, and a history-aware state representation. This project follows that method skeleton directly: a fixed four-phase cycle with safety phases, actions that adjust cycle timing rather than switch phases at each second, a history-aware observation, and comparison against fixed-time baselines. The implementation is single-intersection rather than multi-intersection SUMO, but the control design is intentionally aligned with the cited work.

This paper reports results from a systematic evaluation covering hyperparameter pilot tuning, multi-seed ablations on training duration and network size, and a final difficulty comparison between symmetric and asymmetric-peaky demand conditions. The evaluation framework uses a tuned-static fixed-time baseline as the primary fairness check, ensuring that any reported DQN gain cannot be attributed to a suboptimal fixed-time comparison.

2. Methods

2.1 Environment Design

The traffic signal environment was implemented as a custom Gymnasium environment (Towers et al., 2023) [4]. The environment simulates a single four-approach intersection with four lanes: North, South, East, and West. At each simulation second, vehicles arrive independently in each lane according to a Bernoulli process with time-varying arrival probabilities. The probabilities are modulated by a sinusoidal demand wave to simulate rush-hour-like traffic patterns, with configurable amplitude and period. Vehicle departure capacity is set to two vehicles per second per green lane.

The signal cycle consists of four sequential phases: NS green, NS yellow, EW green, and EW yellow. Base durations are 10 seconds of green and 2 seconds of yellow per direction, yielding a 24-second base cycle. The agent makes exactly one decision per complete cycle, at the cycle boundary. This cycle-level stepping design, following Moreno-Malo et al. (2024) [1], avoids partial-cycle action override and produces cleaner credit assignment compared to second-by-second control.

2.2 Action Space

The agent selects from three discrete actions at each decision step. Action 0 increases the NS green phase by a configurable delta (default 2 seconds) and decreases EW green by the same amount, favoring the NS direction. Action 1 maintains the base cycle unchanged. Action 2 decreases NS green and increases EW green, favoring the EW direction. A minimum green time constraint of 6 seconds is enforced to prevent unsafe signal configurations.

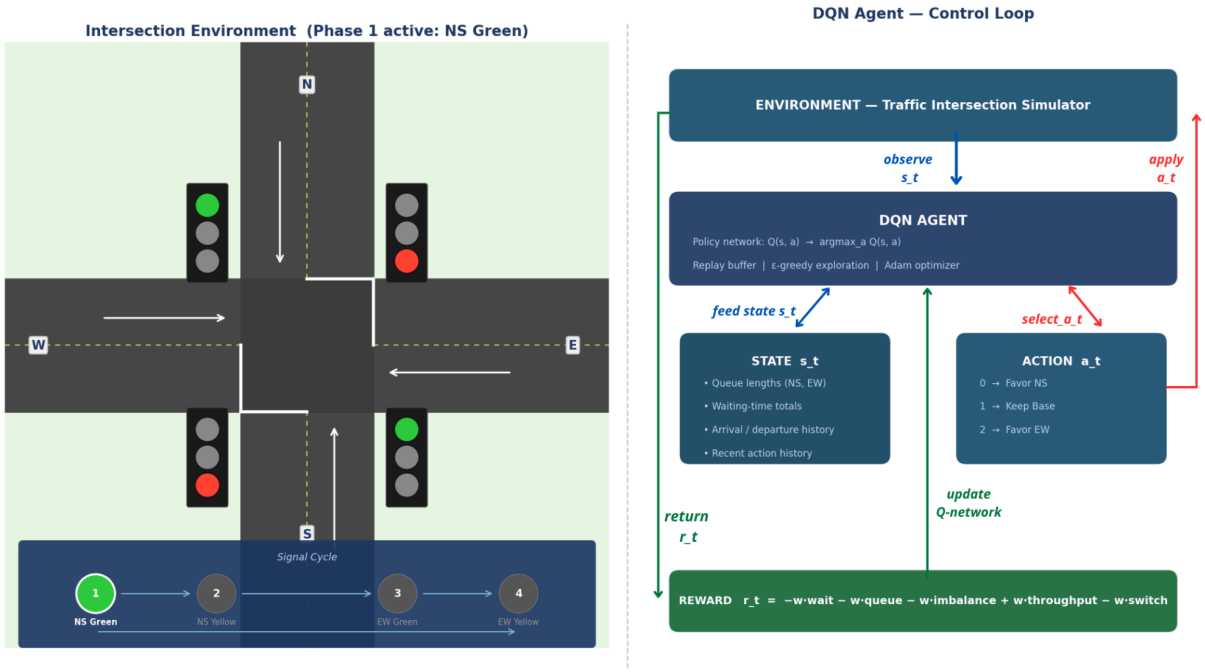


Figure 1: Environment and agent interaction diagram. (Left) Top-down view of the single-intersection environment showing the four-phase signal cycle (NS Green → NS Yellow → EW Green → EW Yellow) with active phase highlighted. (Right) DQN agent control loop: the agent observes the state s_t from the environment, selects an action a_t ∈ {0, 1, 2} via the learned policy Q(s, a), applies it to the environment, and receives a reward r_t used to update the Q-network.

2.3 Observation Space

The 36-dimensional observation vector contains: current queue lengths for all four lanes; cumulative waiting-time totals for all four lanes; a history of per-lane vehicle arrivals over the past K=3 cycles; a history of per-lane vehicle departures over the past K=3 cycles; the current phase identifier; the phase remaining ratio; the last action taken; and the action taken two decision steps prior. All values are clipped and normalized to the range [0, 1] before being passed to the network.

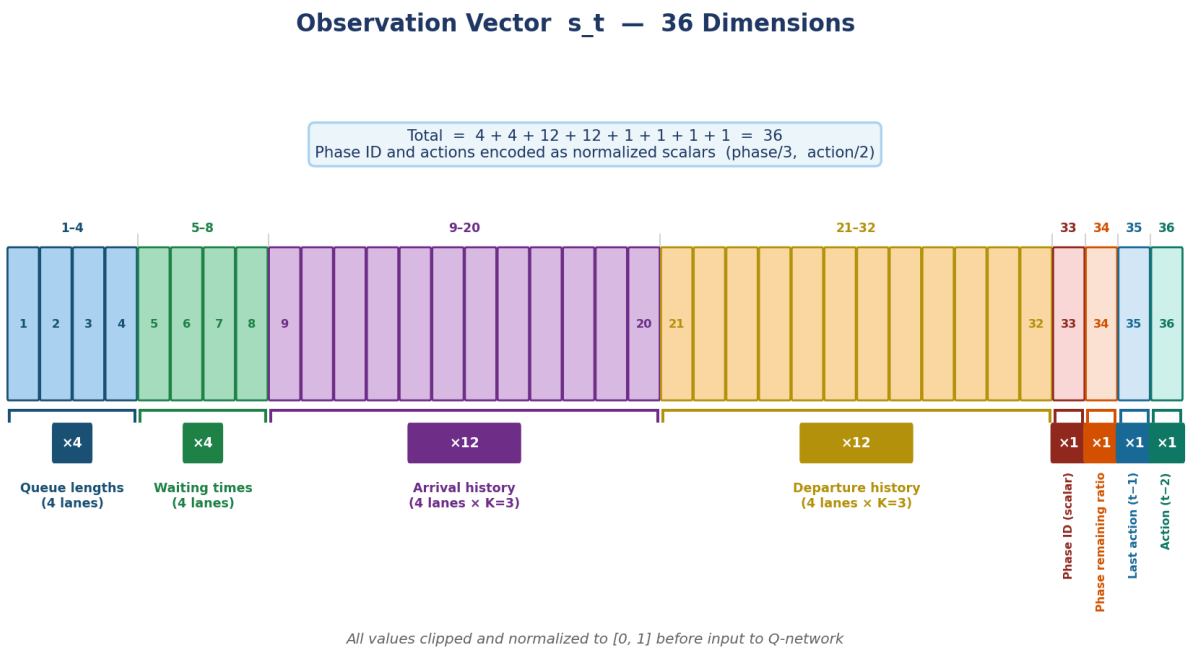


Figure 2: Structure of the 36-dimensional observation vector s_t.

2.4 Reward Function

The cycle-level reward is a weighted linear combination of five components computed over the duration of the most recent cycle. The reward penalizes mean waiting time with weight 1.0, mean queue length with weight 0.2, and mean NS-EW queue imbalance with weight 0.2. It rewards throughput, measured as the number of vehicles departed per cycle second, with weight 0.5, and penalizes unnecessary action changes between consecutive cycles with weight 0.3. Formally:

$$R = -1.0 \cdot wait - 0.2 \cdot queue - 0.2 \cdot imbalance + 0.5 \cdot throughput - 0.3 \cdot action_changed$$

This multi-objective structure encourages the agent to reduce congestion and waiting time while maintaining stable and safe signal operation.

2.5 DQN Agent

The DQN agent maintains an online Q-network and a target Q-network, both implemented as fully connected neural networks with ReLU activations. The architecture used in all final experiments is: input (40) → 1024 → 512 → 64 → output (3 actions). A model-size ablation also evaluates a variant with final hidden dimension 128 instead of 64. The target network is updated by copying the online network weights every 20 episodes. Experience replay uses a replay buffer of 100,000 transitions with a mini-batch size of 64. The optimizer is Adam (Paszke et al., 2019 [3]) with a discount factor of gamma = 0.99.

Exploration follows an epsilon-greedy policy with epsilon decaying linearly from 1.0 to 0.05 over a configurable number of global steps. The agent was trained for either 600 or 1800 episodes depending on the ablation condition, with each episode consisting of 300 decision steps.

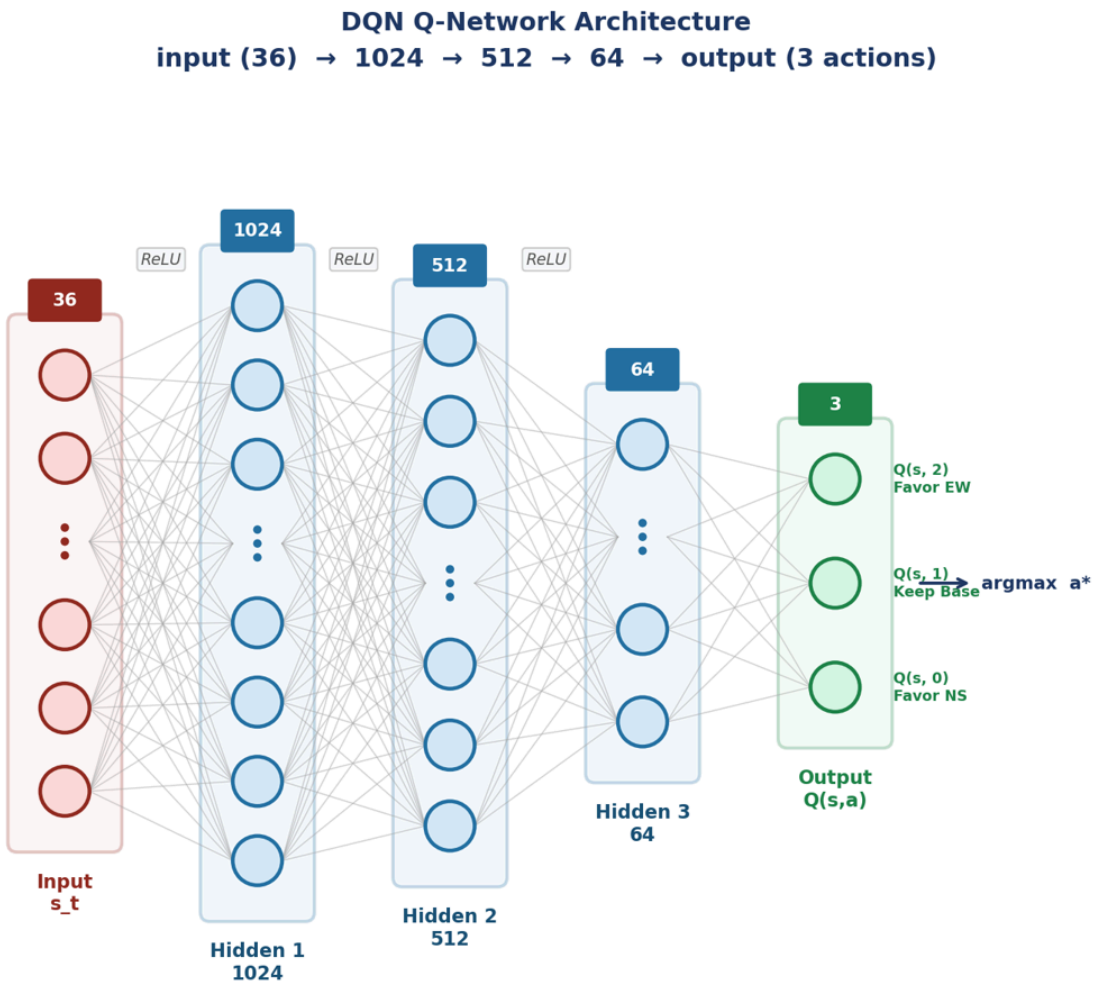


Figure 3: DQN Q-network architecture. The fully connected network takes the 36-dimensional observation vector s_t as input and passes it through three hidden layers of 1024, 512, and 64 units with ReLU activations. The output layer produces three Q-values — one per action — corresponding to favoring NS (action 0), maintaining the base cycle (action 1), and favoring EW (action 2). The selected action a is the argmax over these Q-values. Both the online network and the target network share this architecture; target weights are copied from the online network every 20 episodes.*

2.6 Baseline Policies

Six baseline policies were evaluated for comparison. The Fixed-Time (Keep) policy always selects action 1, maintaining the base cycle without adjustment. The Fixed-Time (Demand-Aware) policy selects a single fixed action based on the relative NS versus EW arrival probabilities at evaluation time, but remains a static policy that does not adapt during the episode. The Fixed-Time (Tuned-Static) policy exhaustively evaluates all three static actions and selects the one yielding the highest average episode reward; this serves as the primary fairness baseline because it represents the best achievable fixed-time performance. Three additional Static policies permanently fix the action to 0, 1, or 2 respectively. The Random Policy samples actions uniformly at random. The Queue-Based Policy compares total NS queue length to total EW queue length and favors the more congested direction, with a deadband of 0.05 to prevent excessive switching.

2.7 Evaluation Metrics

The primary metrics for comparing DQN against the tuned-static baseline are defined as follows. reward_gain_vs_tuned is the difference in average episode reward between DQN and the tuned-static baseline, where higher is better. wait_diff_vs_tuned is the absolute difference in average waiting time per simulation second, computed as the tuned-static wait minus the DQN wait, so that positive values indicate DQN reduces waiting. percent_wait_reduction_vs_tuned is the percentage reduction: (wait_tuned - wait_DQN) / wait_tuned × 100%. All multi-seed results are reported as mean ± standard deviation, with 95% confidence interval half-width equal to 1.96 × std / sqrt(n) shown in parentheses.

2.8 Experimental Protocol

The experimental protocol consists of four stages. First, a single-seed hyperparameter pilot search was conducted on asymmetric-peaky demand, evaluating two learning rates (1e-3 and 3e-4) and two epsilon decay values (360,000 and 600,000 steps). This stage is used only to select a configuration for subsequent stages and is not used as final statistical evidence. Second, multi-seed ablations (n=4, seeds 42–45) examined training duration (600 vs. 1800 episodes) and network size (hidden64 vs. hidden128) under symmetric demand. Third, a final difficulty evaluation (n=5, seeds 42–46) compared the selected configuration under symmetric demand versus asymmetric-peaky demand, with each seed evaluated on 50 episodes. Fourth, an additional symmetric ceiling sweep evaluated six hyperparameter configurations (n=5, seeds 42–46, 100 episodes per seed) to determine whether modest symmetric gains reflect a fundamental ceiling or hyperparameter sensitivity.

3. Results

3.1 Baseline Evaluation (Symmetric Demand, Single Run)

Table 1 presents evaluation results for all policies over 50 episodes under symmetric demand conditions with equal arrival probabilities of 0.30 for all four lanes. The DQN policy achieves the highest average episode reward of -9863.72 and the lowest average wait time of 31.17 seconds, compared to -9944.87 and 31.54 seconds for the best fixed-time baselines. This corresponds to a reward gain of approximately 81 points and a wait time reduction of 0.37 seconds per simulation second. The Queue-Based heuristic outperforms the random policy and the asymmetric static actions but falls short of both the best fixed-time policy and DQN. The

Demand-Aware and Tuned-Static policies both select action 1 under symmetric conditions, producing results identical to Fixed-Time (Keep), confirming that no static action exploits a balanced demand distribution.

Table 1: Full Policy Comparison (50 Episodes, Symmetric Demand). DQN row highlighted.

3.2 Pilot Hyperparameter Tuning (Single Seed, Asymmetric-Peaky)

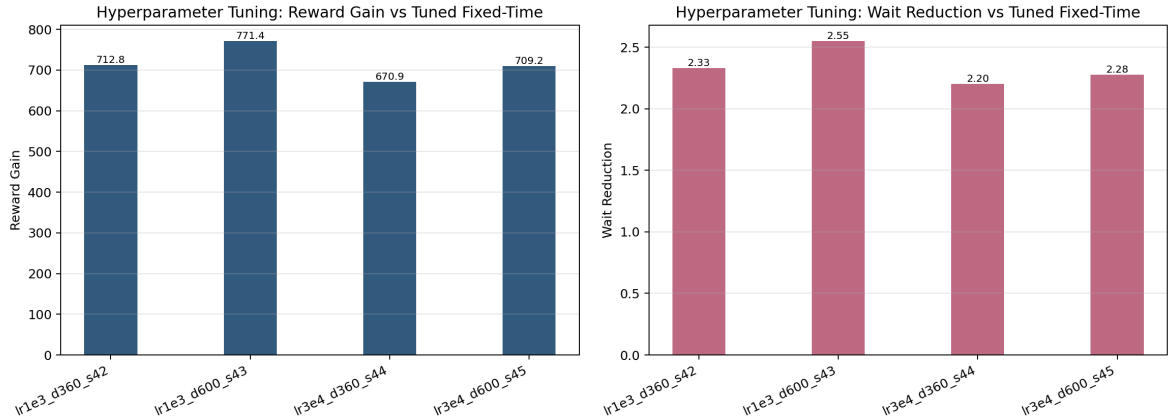
Table 2 and Figure 4 show results of the pilot hyperparameter search on asymmetric-peaky demand. The configuration lr=1e-3 and epsilon_decay=600,000 achieves the highest reward gain of 771.35 and the highest wait reduction of 8.597%, and was selected as the final configuration for all subsequent experiments. The results indicate that a longer exploration schedule (higher epsilon decay) consistently improves performance under asymmetric demand, and that the higher learning rate of 1e-3 outperforms 3e-4 across both decay values tested.

Policy	Avg Reward	Avg Wait	Avg Queue	Avg Departed	Avg Switches
Fixed-Time (Keep)	-9944.87	31.5431	5.9916	8649.0	0.0
Fixed-Time (Demand-Aware)	-9944.87	31.5431	5.9916	8649.0	0.0
Fixed-Time (Tuned-Static)	-9944.87	31.5431	5.9916	8649.0	0.0
Fixed-Time (Static a=0)	-10509.38	33.3745	6.1229	8650.1	1.0
Fixed-Time (Static a=1)	-9944.87	31.5431	5.9916	8649.0	0.0
Fixed-Time (Static a=2)	-10485.71	33.2987	6.1106	8647.8	1.0
Random Policy	-10393.07	32.8015	6.0824	8649.0	201.0
Queue-Based Policy	-10277.71	32.5231	6.0440	8649.7	110.3
DQN Policy	**-9863.72**	**31.1727**	**5.9066**	**8649.3**	**133.1**

Table 2: Pilot Hyperparameter Search (Single Seed, Asymmetric-Peaky Demand)

LR / Eps Decay	Reward Gain	Wait Diff	Wait Reduction (%)
lr=1e-3, decay=360000	712.85	2.332	7.861%
lr=1e-3, decay=600000	771.35	2.551	8.597%
lr=3e-4, decay=360000	670.88	2.204	7.427%
lr=3e-4, decay=600000	709.24	2.276	7.670%

Figure 4: Hyperparameter tuning comparison — reward gain and wait reduction vs. tuned fixed-time



3.3

Ablation Study: Training Duration (n=4 Seeds)

Table 3 presents the duration ablation results averaged over four seeds under symmetric demand. Training for 1800 episodes yields a marginally higher mean reward gain (60.14 vs. 52.42) and wait reduction (0.827% vs. 0.793%) compared to 600 episodes. However, the 1800-episode condition shows substantially higher variance across seeds (std of 87.23 vs. 25.46 in reward gain), making the difference statistically unreliable. This suggests that under

symmetric demand, additional training provides limited mean benefit and introduces greater instability.

Table 3: Duration Ablation Results (mean ± std, 95% CI in parentheses, n=4).

Config	Reward Gain vs Tuned	Wait Diff vs Tuned	Wait Reduction (%)
ep600	52.42 ± 25.46 (±24.95)	0.249 ± 0.101 (±0.099)	0.793%
ep1800	60.14 ± 87.23 (±85.49)	0.260 ± 0.311 (±0.305)	0.827%

3.4 Ablation Study: Model Size (n=4 Seeds)

Table 4 presents the model-size ablation results under symmetric demand. The hidden64 configuration substantially outperforms hidden128 in both mean performance and training stability. The hidden128 model shows extreme inter-seed variance (std of 138.43 in reward gain and 0.473 in wait difference), indicating that the larger output-layer bottleneck is prone to unstable training in this environment. The hidden64 architecture is therefore the recommended configuration for this observation space and action dimensionality.

Table 4: Model Size Ablation Results (mean ± std, 95% CI in parentheses, n=4)

Config	Reward Gain vs Tuned	Wait Diff vs Tuned	Wait Reduction (%)
hidden64	106.62 ± 44.23 (±43.35)	0.424 ± 0.169 (±0.165)	1.347%
hidden128	1.94 ± 138.43 (±135.66)	0.076 ± 0.473 (±0.464)	0.240%

3.5 Final Results: Symmetric vs. Asymmetric-Peaky Demand (n=5 Seeds)

Table 5 and Figure 5 present the final main results comparing symmetric and asymmetric-peaky demand conditions using the selected configuration (hidden_dim=64, lr=1e-3, epsilon_decay=600,000), averaged over five seeds. Under symmetric demand, DQN achieves a reward gain of 115.87 and a wait reduction of 1.455%. Under asymmetric-peaky demand, the reward gain increases to 706.00 and the wait reduction increases to 7.741%. In both scenarios the tuned-static baseline selects action 1 (avg_tuned_action=1.00), confirming that no static action can match DQN performance and that all DQN gains arise from dynamic cycle-level adaptation rather than from identifying a better fixed action. Figure 4 shows the policy-level comparison on the best tuned run, illustrating DQN's advantage in both reward and average wait time across all baselines.

The large gap between the two conditions — a roughly five-fold difference in wait reduction — has a direct structural explanation. Under symmetric demand, all four lanes receive equal average traffic volume (arrival probability 0.30 each). In this setting, a fixed-time policy that splits green time equally between NS and EW is already near-optimal because there is no persistent directional imbalance for the agent to exploit. The DQN can make marginal cycle-to-cycle improvements, but the headroom above the best fixed-time policy is inherently small. Under asymmetric-peaky demand, by contrast, the NS and EW approaches receive systematically different traffic volumes (0.40 and 0.20 for NS, 0.35 and 0.15 for EW), and demand fluctuates with higher amplitude and a shorter period. A fixed-time policy cannot respond to this imbalance and consistently under-serves the congested direction. The DQN learns to dynamically shift green time toward whichever direction is currently more congested,

providing a structural advantage that compounds over each cycle. This is precisely the scenario where adaptive signal control is most valuable, and the results confirm this directly.

Table 5: Final Results by Demand Condition (mean ± std, 95% CI in parentheses, n=5)

Scenario	Reward Gain vs Tuned	Wait Diff vs Tuned	Wait Reduction (%)	Tuned Action
Symmetric	115.87 ± 56.92 (±49.89)	0.458 ± 0.227 (±0.199)	1.455%	1.00
Asymmetric-Peaky	706.00 ± 49.85 (±43.70)	2.292 ± 0.182 (±0.159)	7.741%	1.00

Figure 4: Main difficulty results — symmetric vs. asymmetric-peaky demand

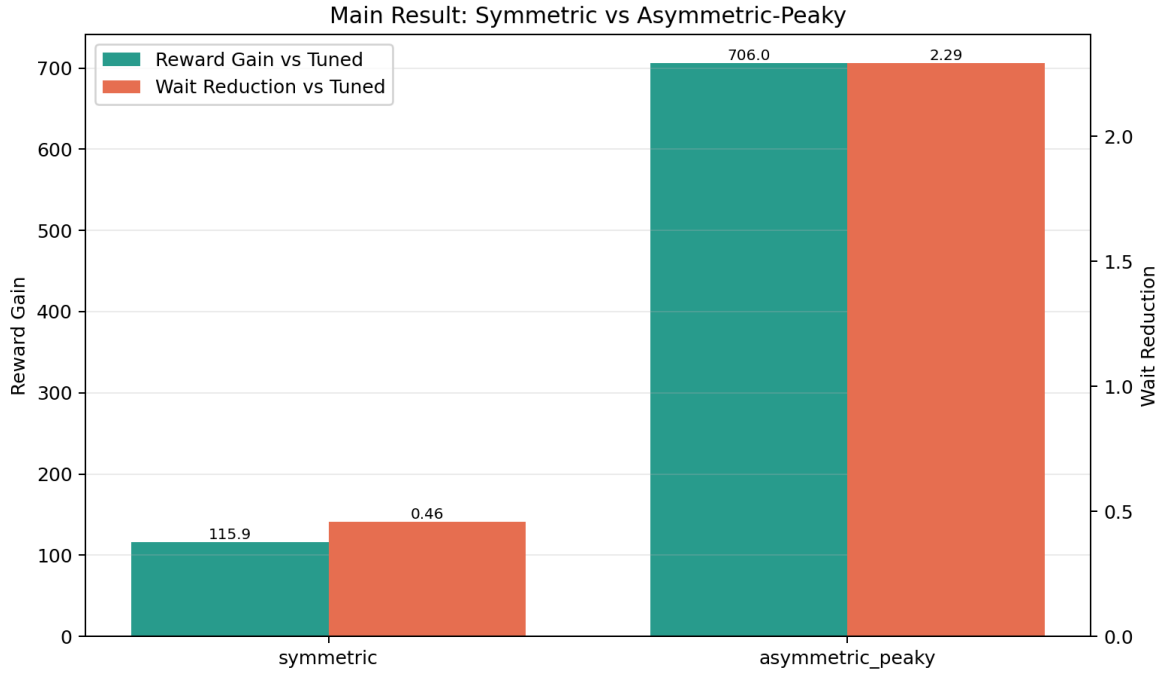
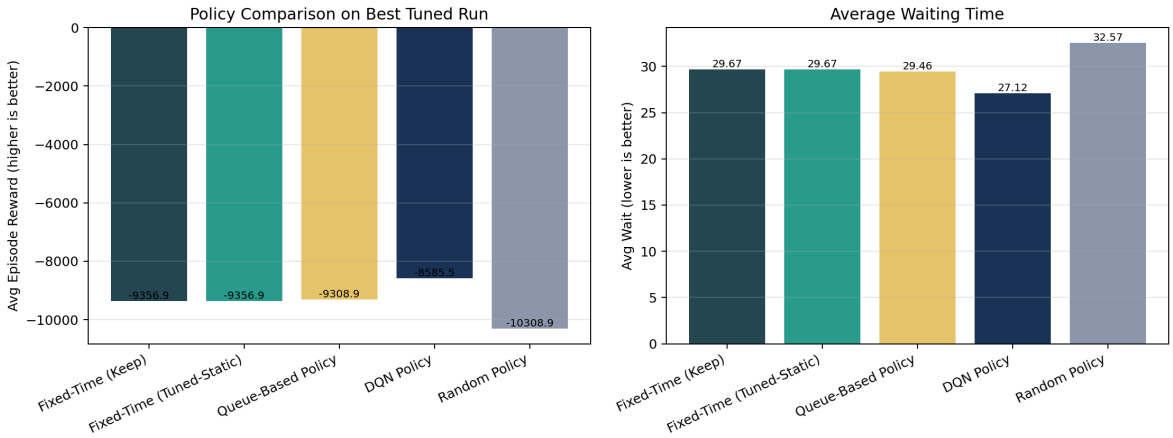


Figure 5: Policy comparison on best tuned run — reward and average wait.



3.6 Symmetric Ceiling Diagnostics (n=5 Seeds)

To investigate whether modest symmetric gains reflect a fundamental ceiling or hyperparameter sensitivity, an additional sweep evaluated six configurations under symmetric demand with n=5 seeds and 100 evaluation episodes per seed. The six settings varied epsilon decay (180,000, 240,000, or 600,000 steps) and the action-switch penalty weight (0.05, 0.15, or 0.30). Table 6 presents the results.

Table 6: Symmetric Ceiling Sweep (mean ± std, 95% CI in parentheses, n=5).

Setting	Reward Gain vs Tuned	Wait Diff vs Tuned	Wait Reduction (%)	Avg Switch Count
d240_sw015	105.84 ± 53.51 (±46.90)	0.365 ± 0.185 (±0.163)	1.158%	90.94
base	99.45 ± 55.50 (±48.64)	0.409 ± 0.210 (±0.184)	1.297%	113.75
d180_sw015	88.31 ± 53.34 (±46.76)	0.302 ± 0.182 (±0.159)	0.957%	102.45
d180	42.26 ± 51.56 (±45.19)	0.189 ± 0.188 (±0.165)	0.600%	71.79
d180_sw005	-23.15 ± 234.15 (±205.24)	-0.078 ± 0.759 (±0.665)	-0.248%	77.54
d240	-24.42 ± 248.96 (±218.22)	-0.014 ± 0.835 (±0.732)	-0.044%	89.40

The best-performing symmetric configurations (d240_sw015 and base) yield wait reductions of approximately 1.0% to 1.3%, substantially smaller than the 7.741% observed under asymmetric-peaky demand. Two configurations (d240 and d180_sw005) produce negative reward gains and near-zero or slightly negative wait differences, indicating that these hyperparameter choices are unstable and can cause DQN to underperform the tuned-static baseline. This sweep supports two conclusions: symmetric demand is close to a performance ceiling for this control structure, but hyperparameter stability still matters and some settings reliably achieve small positive gains while others do not.

4. Conclusions

This project implemented and evaluated a cycle-level DQN controller for single-intersection traffic signal control, aligned with the method skeleton of Moreno-Malo et al. (2024) [1]. The agent consistently outperforms all evaluated baseline policies across both symmetric and asymmetric demand conditions, with the magnitude of improvement scaling strongly with the degree of demand asymmetry.

The contrast between symmetric and asymmetric results reflects a fundamental property of adaptive signal control. Under symmetric demand, all lanes receive equal average traffic, so a balanced fixed-time policy is already close to optimal and leaves little room for improvement. The DQN still achieves small positive gains by responding to short-term fluctuations, but the ceiling is low. Under asymmetric-peaky demand, persistent directional imbalance means fixed-time policies consistently waste green time on less congested approaches. The DQN learns to reallocate green time dynamically toward whichever direction is currently congested, compounding savings across every cycle. This is where adaptive control provides the most value,

and the roughly five-fold difference in wait reduction between the two conditions — 1.455% versus 7.741% — directly quantifies that advantage. The implication for deployment is clear: cycle-level DQN control is most beneficial in intersections with unequal or time-varying directional demand, and its advantage diminishes in settings that closely resemble the symmetric, balanced assumption underlying fixed-time design.

The primary limitations of this work are threefold. First, the simulator is a custom single-intersection environment rather than a high-fidelity multi-intersection platform such as SUMO. The control structure is intentionally designed for compatibility with future SUMO migration, but current results apply only to the single-intersection setting. Second, the pilot hyperparameter search is conducted on a single seed, so tuning conclusions are directional rather than statistically definitive. Third, some ablation conditions, particularly ep1800 and hidden128, exhibit high inter-seed variance that warrants further investigation.

The strongest result in this study is observed under asymmetric-peaky demand: DQN improves reward by approximately 706 points and reduces the average waiting time metric by 2.292 units, corresponding to a 7.741% reduction versus the tuned-static fixed-time baseline. Under symmetric demand, additional ceiling diagnostics confirm that gains of approximately 1.0% to 1.3% in wait reduction are achievable under stable hyperparameter settings, consistent with a near-ceiling regime where there is little directional imbalance for the agent to exploit.

Future work could extend this implementation to a multi-intersection SUMO environment, explore Double DQN or Dueling DQN architectures to address Q-value overestimation, and apply the method to real-world traffic count data rather than synthetic stochastic arrivals.

5. Works Cited

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